Groww API

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Backtesting

Fetch historical candle data and instrument information for backtesting trading strategies using Groww APIs

> *"Note: Currently, Backtesting APIs only support CASH and FNO segments."*

Groww Symbol

Groww symbol is a easy to construct unique identifier for an instrument across exchanges and segments. It is formed by concatenating

- **Exchange** - Where the instrument is traded
- **Trading Symbol** - The name/ticker of the instrument
- **Expiry Date** - Only for derivatives (format: DDMmmYY, example: 23Jan25)
- **Strike Price** - Only for options (the target price level)
- **Option Type** - Only for derivatives:
  - CE = Call Option
  - PE = Put Option
  - FUT = Futures

For Stocks and Indices: Only exchange and trading symbol are used.

For Futures: Exchange, trading symbol, expiry date, and "FUT" are used.

For Options: All components are used including strike price and option type (CE/PE).

For example:

- Equity: `NSE-WIPRO`, `BSE-RELIANCE`

- Index: `NSE-NIFTY`, `BSE-SENSEX`
- Future: `NSE-NIFTY-30Sep25-FUT`, `BSE-SENSEX-25Sep25-FUT`
- Call Option: `NSE-NIFTY-30Sep25-24650-CE`, `BSE-SENSEX-25Sep25-79500-CE`
- Put Option: `NSE-NIFTY-30Sep25-24650-PE`, `BSE-SENSEX-25Sep25-79500-PE`

Groww symbol also exists in the instruments csv file and it can be obtained from the Get Instruments API.

Get Expiries

This API retrieves available expiry dates for derivatives instruments (FNO) for a given exchange and underlying symbol. Useful for backtesting options and futures strategies. Data of FNO instruments are available from 2020.

Python SDK Usage

```python
from growwapi import GrowwAPI

# Groww API Credentials (Replace with your actual credentials)
API_AUTH_TOKEN = "your_token"

# Initialize Groww API
groww = GrowwAPI(API_AUTH_TOKEN)

expiries_response = groww.get_expiries(
    exchange=groww.EXCHANGE_NSE,
    underlying_symbol="NIFTY",
    year=2024,
    month=1
)
print(expiries_response)
```

Request Schema

Name	Type	Description
exchange *	string	Stock exchange (NSE, BSE)

Name	Type	Description
underlying_symbol *	string	Underlying symbol for which expiry dates are required (e.g., NIFTY, BANKNIFTY, RELIANCE)
year	integer	Year for which expiry dates are required (2020 - current year). If year is not specified, current year is considered.
month	integer	Month for which expiry dates are required (1-12). If month is not specified, expiries of the entire year is returned.

* required parameters

Response Payload

```
{
    "expiries" : [
        "2024-01-25",
        "2024-01-31",
        "2024-02-29",
        "2024-03-28"
    ]
}
```

Response Schema

Name	Type	Description
expiries	array[string]	Array of expiry dates in YYYY-MM-DD format

Get Contracts

This API retrieves available contract symbols for derivatives instruments for a given exchange, underlying symbol, and expiry date. Essential for backtesting specific options or futures contracts by passing them in Candles API. Data of FNO instruments are available from 2020.

Python SDK Usage

```python
from growwapi import GrowwAPI

# Groww API Credentials (Replace with your actual credentials)
API_AUTH_TOKEN = "your_token"

# Initialize Groww API
groww = GrowwAPI(API_AUTH_TOKEN)

contracts_response = groww.get_contracts(
    exchange=groww.EXCHANGE_NSE,
    underlying_symbol="NIFTY",
    expiry_date="2025-01-25"
)
print(contracts_response)
```

Request Schema

Name	Type	Description
exchange *	string	Stock exchange (NSE, BSE)
underlying_symbol *	string	Underlying symbol for which contracts are required (1-20 characters)
expiry_date *	string	Expiry date in YYYY-MM-DD format for which contracts are required

* required parameters

Response Payload

```
{
    "contracts": [
        "NSE-NIFTY-02Jan25-28500-PE",
        "NSE-NIFTY-02Jan25-24000-PE",
        "NSE-NIFTY-02Jan25-26800-PE",
        "NSE-NIFTY-02Jan25-27450-PE",
        "NSE-NIFTY-02Jan25-19050-PE",
        "NSE-NIFTY-02Jan25-22300-PE",
        "NSE-NIFTY-02Jan25-28150-CE"
    ]
}
```

Response Schema

Name	Type	Description
contracts	array[string]	Array of groww symbols of the contracts available for the given expiry date

Get Historical Candles

Fetch historical candle data for backtesting trading strategies. This API provides

- Historical OHLC (Open, High, Low, Close) data for all instruments
- Volume for tradable instruments (Equities and FNO)
- Open Interest (OI) for FNO

Data of Equities, Indices and FNO instruments are available from 2020.

Python SDK Usage

```
from growwapi import GrowwAPI

# Groww API Credentials (Replace with your actual credentials)
API_AUTH_TOKEN = "your_token"

# Initialize Groww API
groww = GrowwAPI(API_AUTH_TOKEN)

historical_candles_response = groww.get_historical_candles(
```

```
    exchange=groww.EXCHANGE_NSE,
    segment=groww.SEGMENT_CASH,
    groww_symbol="NSE-WIPRO",
    start_time="2025-09-24 10:56:00",
    end_time="2025-09-24 12:00:00",
    candle_interval=groww.CANDLE_INTERVAL_MIN_30
)
print(historical_candles_response)


# OR
# You can also use expiries and contracts API to get historical data of FNO instrument

jan2024_nifty_expiries = groww.get_expiries(
    exchange=groww.EXCHANGE_NSE,
    underlying_symbol="NIFTY",
    year=2024,
    month=1
)
```

Request Schema

Name	Type	Description
exchange *	string	Stock exchange
segment *	string	Segment of the instrument such as CASH, FNO etc.
groww_symbol *	string	Groww symbol of the instrument for which historical data is required
start_time *	string	Start time in yyyy-MM-dd HH:mm:ss or epoch seconds format from which data is required
end_time *	string	End time in yyyy-MM-dd HH:mm:ss or epoch seconds format until which data is required
candle_interval *	string	Interval for which data is required

* required parameters

Response Payload

All prices in rupees.

```
{
  "candles": [
      [
          "2025-09-24T10:30:00", // candle timestamp in yyyy-MM-dd HH:mm:ss format
          245.95, // open price
          246.15, // high price
          245.05, // low price
          245.6,  // close price
          735060, // volume
          null // open interest (only for FNO instruments, null for others)
      ],
      [
          "2025-09-24T11:00:00",
          245.64,
          245.66,
          244.8,
          244.94,
          682373,
          null
      ],
      [
          "2025-09-24T11:30:00",
          244.95,
          245.28,
          244.6,
          245.13,
          353800,
          null
      ],
```

Response Schema

Name	Type	Description
candles	array[array]	Array of candle data. Each candle contains: timestamp (yyyy-MM-dd HH:mm:ss), open, high, low, close, volume, open interest
closing_price	float	Closing price of the instrument
start_time	string	Start time in yyyy-MM-dd HH:mm:ss format
end_time	string	End time in yyyy-MM-dd HH:mm:ss format
interval_in_minutes	int	Interval in minutes

Backtesting Data Limits

Candle Intervals	Max Duration per Request
1 min, 2 min, 3 min, 5 min	30 days
10 min, 15 min, 30 min	90 days
1 hour, 4 hours, 1 day, 1 week, 1 month	180 days

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